

CO3 ASSESSMENT TOOL 2 – CLASS TEST

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
C PROGRAMMING

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CO3 - AT2 - Class Test

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QUESTIONS

Q1
CO3 - AT2 - Class Test
20 marks

Q2
CO3 - AT2 - Class Test
20 marks

Q3
CO3 - AT2 - Class Test
20 marks

Q4
CO3 - AT2 - Class Test
20 marks

Q5
CO3 - AT2 - Class Test
20 marks

5/5 answered

Q1 CO3 - AT2 - Class Test

20 marks

1. Write a program to cyclically rotate an array by one position to the right.
Sample Input: 1 2 3 4 5
Sample Output: 5 1 2 3 4

Your Code

```
#include <stdio.h>
2
3 int main() {
4     int n;
5     scanf("%d", &n);
6
7     int a[n];
8     for (int i = 0; i < n; i++) {
9         scanf("%d", &a[i]);
10    }
11
12    int last = a[n - 1];
13    for (int i = n - 1; i >= 1; i--) {
14        a[i] = a[i - 1];
15    }
16    a[0] = last;
17
18    for (int i = 0; i < n; i++) {
19        printf("%d ", a[i]);
20        if (i < n - 1) printf(" ");
21    }
22    return 0;
23 }
```

Run Save

Input

5
1 2 3 4 5

Output

Test Case 1

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Q1
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20 marks

Q2
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20 marks

Q3
CO3 - AT2 - Class Test
20 marks

Q4
CO3 - AT2 - Class Test
20 marks

Q5
CO3 - AT2 - Class Test
20 marks

5/5 answered

Q2 CO3 - AT2 - Class Test

20 marks

2. Write a program to find the String Palindrome.
Sample Input: MADAM
Sample Output: It is Palindrome

Your Code

```
#include <stdio.h>
#include <string.h>
3
4 int main() {
5     char s[100];
6     int i, j, isPalin = 1;
7
8     scanf("%s", s);
9
10    i = 0;
11    j = (int)strlen(s) - 1;
12
13    while (i < j) {
14        if (s[i] != s[j]) {
15            isPalin = 0;
16            break;
17        }
18        i++;
19        j--;
20    }
21
22    if (isPalin)
23        printf("It is Palindrome");
24    else
25        printf("It is Not Palindrome");
26
27    return 0;
28 }
```

Run Save

Input

MADAM

Output

Test Case 1

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QUESTIONS

- Q1
CO3 - AT2 - Class Test
20 marks
- Q2
CO3 - AT2 - Class Test
20 marks
- Q3
CO3 - AT2 - Class Test
20 marks
- Q4
CO3 - AT2 - Class Test
20 marks
- Q5
CO3 - AT2 - Class Test
20 marks

3/5 answered

Q3 CO3 - AT2 - Class Test

20 marks

3. Write a program to find the maximum element in each row of a matrix.

Your Code

```
#include <stdio.h>

int main() {
    int rows, cols, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    int matrix[rows][cols];

    printf("Enter elements of the matrix:\n");
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    printf("Maximum element in each row:\n");
    for (i = 0; i < rows; i++) {
        int max = matrix[i][0];
        for (j = 1; j < cols; j++) {
            if (matrix[i][j] > max) {
                max = matrix[i][j];
            }
        }
        printf("Row %d: %d\n", i + 1, max);
    }

    return 0;
}
```

Run

Save

Input

```
3 3
1 2 3
4 6 -2
5 3 7
```

Output

CO3 - AT2 - Class Test

← Back

QUESTIONS

- Q1
CO3 - AT2 - Class Test
20 marks
- Q2
CO3 - AT2 - Class Test
20 marks
- Q3
CO3 - AT2 - Class Test
20 marks
- Q4
CO3 - AT2 - Class Test
20 marks
- Q5
CO3 - AT2 - Class Test
20 marks

3/5 answered

Q4 CO3 - AT2 - Class Test

20 marks

4. Write a program to find the transpose of a matrix.

Sample Input:

```
1 2 3
4 5 6
7 8 9
```

Sample Output:

```
1 4 7
2 5 8
3 6 9
```

Your Code

```
#include <stdio.h>

int main() {
    int rows = 3, cols = 3;
    int matrix[3][3];
    int transpose[3][3];

    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            transpose[j][i] = matrix[i][j];
        }
    }

    for (int i = 0; i < cols; i++) {
        for (int j = 0; j < rows; j++) {
            printf("%d ", transpose[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Run

Save

Input

```
1 2 3
4 5 6
7 8 9
```

Output

Visible Test Cases

Test Case 1

CO3 - AT2 - Class Test

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QUESTIONS

- Q1
CO3 - AT2 - Class Test
20 marks
- Q2
CO3 - AT2 - Class Test
20 marks
- Q3
CO3 - AT2 - Class Test
20 marks
- Q4
CO3 - AT2 - Class Test
20 marks
- Q5
CO3 - AT2 - Class Test
20 marks

3/5 answered

Q5 CO3 - AT2 - Class Test

20 marks

5. Write a program to find and print duplicate elements in an array.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j;
5
6     printf("Enter number of elements: ");
7     scanf("%d", &n);
8
9     int arr[n];
10
11     printf("Enter %d elements\n", n);
12     for (i = 0; i < n; i++) {
13         scanf("%d", &arr[i]);
14     }
15
16     printf("Duplicate elements in the array\n");
17     int found = 0;
18
19     for (i = 0; i < n; i++) {
20         for (j = i + 1; j < n; j++) {
21             if (arr[i] == arr[j]) {
22                 printf("%d\n", arr[i]);
23                 found = 1;
24                 break;
25             }
26         }
27     }
28
29     if (!found) {
30         printf("No duplicates found\n");
31     }
32
33     return 0;
34 }
```

Run

Save

Input

6
1 2 3 2 4 5

Output